1 The dual solution, numerical examples

In this section we will discuss the numerical computation of the linearised dual problem and present some results. We will treat the following non-linear equations from fluid mechanics:

- The Burgers equation in one space dimension
- The Euler equations from gas dynamics in one and two space dimensions
- A hyperbolic model problem with relaxation in one space dimension
- A two-phase problem in one space dimension.

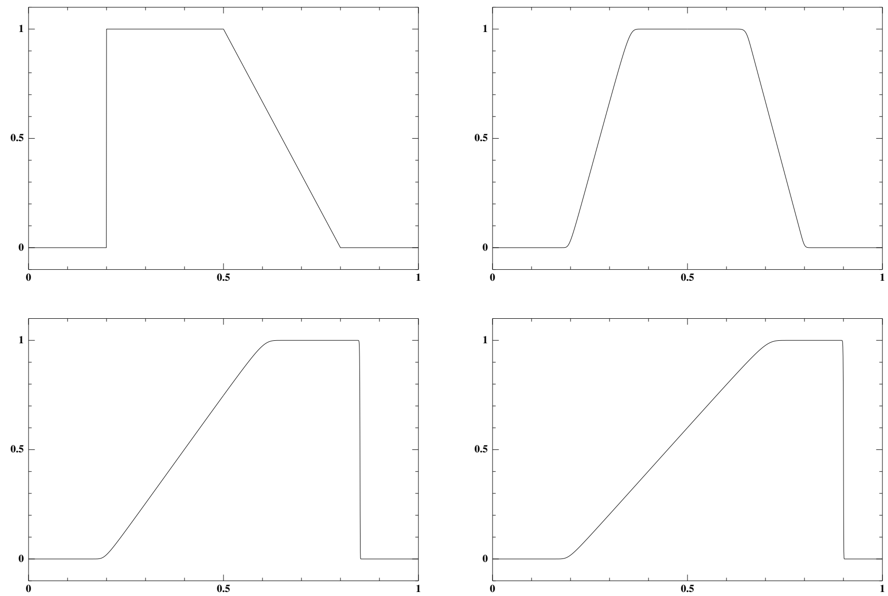
As mentioned in the analysis the linearised dual problem corresponding to a hyperbolic problem takes the following form:

$$\begin{aligned} -\varphi_t - \tilde{A}(u,U)\varphi_x + \tilde{B}(u,U)\varphi - \epsilon\varphi_{xx} &= 0, \quad \text{in } \mathbb{R}\times(0,T) && (1)\\ \phi(\cdot,T) &= \psi, \qquad \text{in } \mathbb{R}. && (2) \end{aligned}$$

Where ψ is chosen in a way corresponding to the type of error control expected. For instance the choice $\psi = \frac{e}{\|e\|_{L_2(\Omega)}}$ corresponds to error control in the $_\infty(L_2)$ norm. Unfortunately the exact solution appears in the dual-problem, in the operator as well as in ψ (at least for some choices of ψ). In the numerical approach typically the exact solution is replaced by a finite element solution on a finer grid thus giving an approximation to the dual operator as well as to the error. In this note we will present some justification of this approach by comparing the results with those obtained using the exact operator and the exact error, for the two first examples where analytical solutions are known in the inviscid case. We have also computed the dual solution using only the coarse finite element solution in the dual operator.

1.1 The Burgers equation

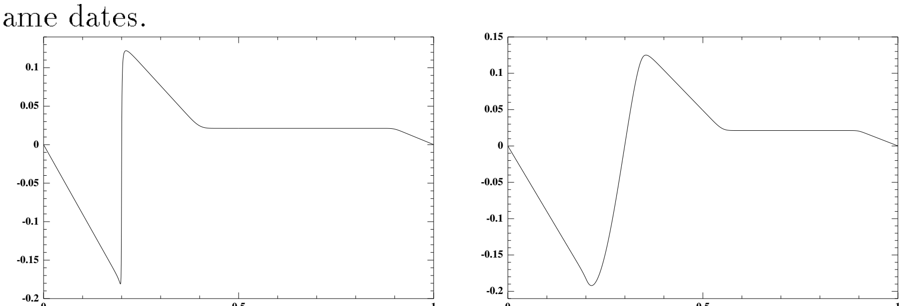
We have studied a case with a rarefaction wave and a shock. The solutions take the following form at t=0, t=0.3 T, t=0.8 T, t=T, with the inviscid, analytical solution dotted.

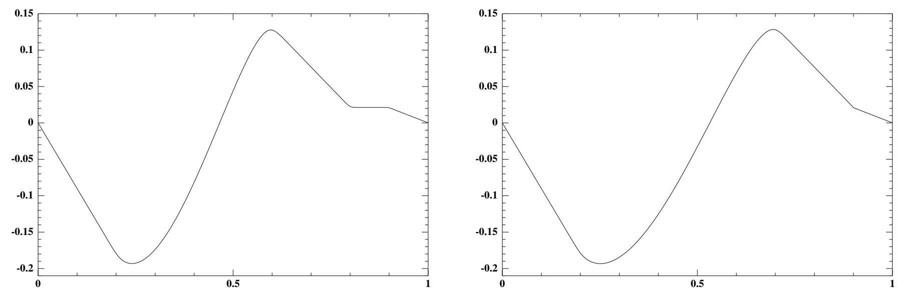


For this particular solution we have solved the dual problem using the following approximations of the dual operator $A(u, U)$ and of the error e.

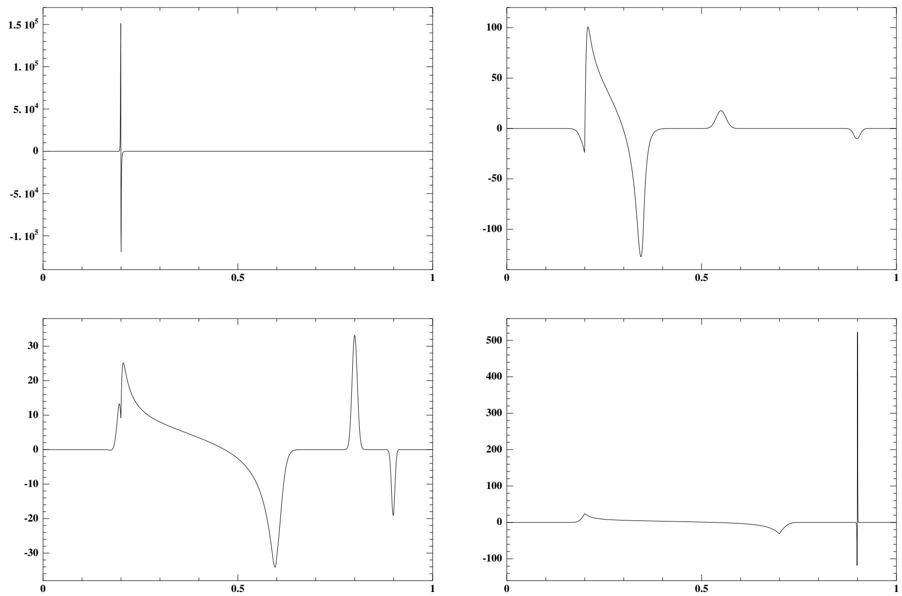
- $A(u, U)$, $e = u - U$, where u is the analytical, inviscid solution and U is the finite element solution.
- $A(U, U)$, $e = u - U$, where u is the analytical, inviscid solution and U is the finite element solution.
- $A(\tilde{u}, U)$, $e = \tilde{u} - U$, where $\tilde{u}$ U are finite element solution on differentmeshes $(h/2,\ h)$.

The dualsolutions for the first two cases takes the following form at the same dates.

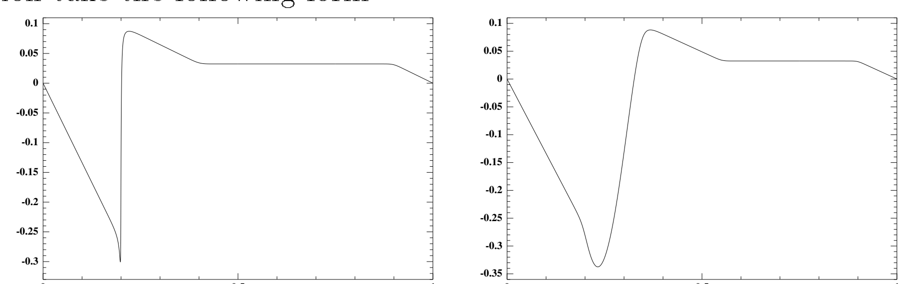


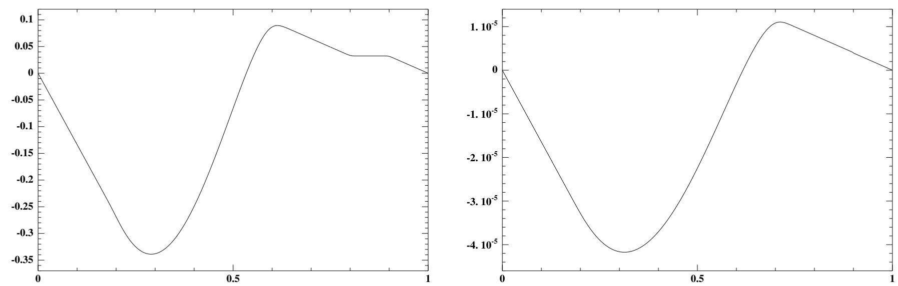


with the corresponding second derivatives:

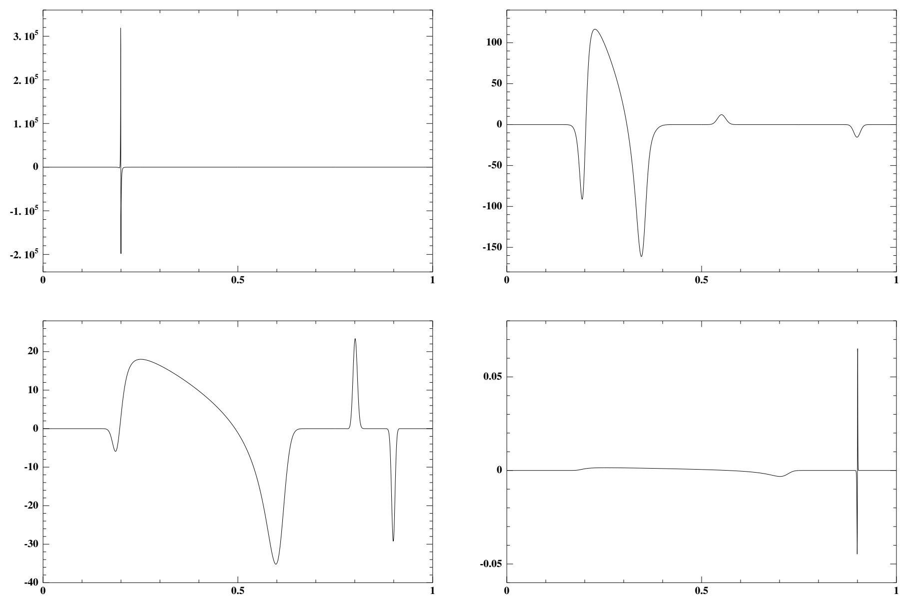


and the dual solution using only finite element solutions for the approximation take the following form





with the corresponding second derivatives:



In the following graph we plot the stability factors,

$$S_2 = \int_0^T \|\epsilon^{1/2}\varphi_x x(\cdot,t)\|_{L_2(\Omega)} dt$$

for the three different approaches using three different values on the viscosity ($\epsilon = 0.0001$, $\epsilon = 0.00005$, $\epsilon = 0.00001$)

1.2 The Euler equations in one space dimension

The dualproblem for the Euler equations in one space dimension was studied numerically in [**?**] as mentioned above we will complement those results with a study of the effects of perturbations on the dual operator on the stability factors. We solve the Riemann problem suggested by Sod in the cases presented above.

1.3 hyperbolic relaxation problem

The following system of equation was introduced in [?] as a model problem for two-phase flow.

$$\partial_t u_1 + \partial_x \left(\frac{u_1^2}{2}\right) - \epsilon_1 \partial_{xx} u_1 = \frac{1}{\tau}(u_2 - u_1),\ x \in \mathbb{R}, t > 0 \tag{3}$$

$$\partial_t u_2 + c\partial_x \left(\frac{u_2^2}{2}\right) - \epsilon_2 \partial_{xx} u_2 = \frac{1}{\tau}(u_2 - u_1),\ x \in \mathbb{R}, t > 0 \tag{4}$$

$$u_i(x,0) = u_i^0(x),\ x \in \mathbb{R},\ i = 1,2. \tag{5}$$

Here c is a constant, much less than one giving u_2 the inert character of the liquid, u_1 and u_2 represent the velocities of the gas and of the liquid, respectively, and τ is a small positive parameter indicating the relaxation time of the system, as well as playing the role of droplet radius. This system is strictly hyperbolic with relaxation. In fact it relates to the complete (one-dimensional) two-phase model much in the same way as the Burgers equation relates to the Euler equations. We have solved the dual problem corresponding to the following forward solution,

for different values of the relaxation parameter tau

1.4 two-phase problem